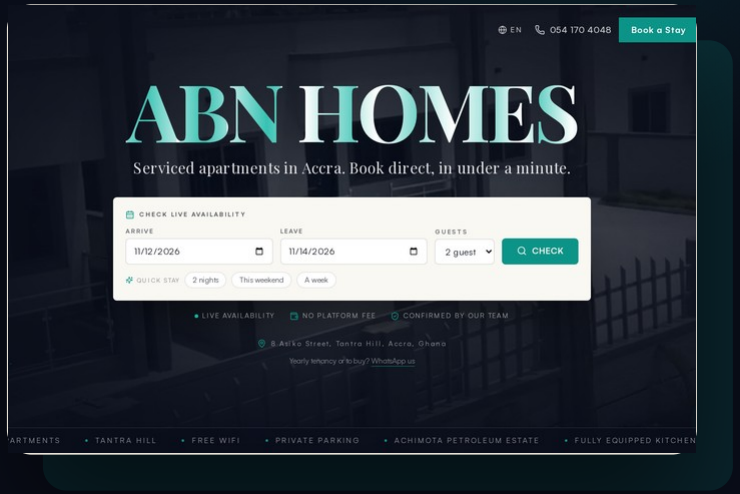
PRODUCT WALKTHROUGH

ABN Homes

How the platform works — from first visit to owner payout

abnhomes.vercel.app · live production · November 2026

- Guest journey
- Staff console
- Owner & tenant portals



ONE PLATFORM

Four audiences, one source of truth

ABN Homes runs short-stay apartments and long-let leases in Accra. The platform replaces spreadsheets and WhatsApp threads: inventory, bookings, money, operations and reporting live in one accountable system with a double-entry ledger the database itself protects.

Guests

browse, book, pay, check in

Staff

22-page console for daily operations

Owners

statements, expenses, payouts

Tenants

lease, rent account, requests

42
database tables

22
admin pages

15
screenshots flows verified

37
tests passing

01 / 03 SECTION

The Guest Experience

Marketing site, marketplace, booking, payment — the public-facing loop, all server-validated.

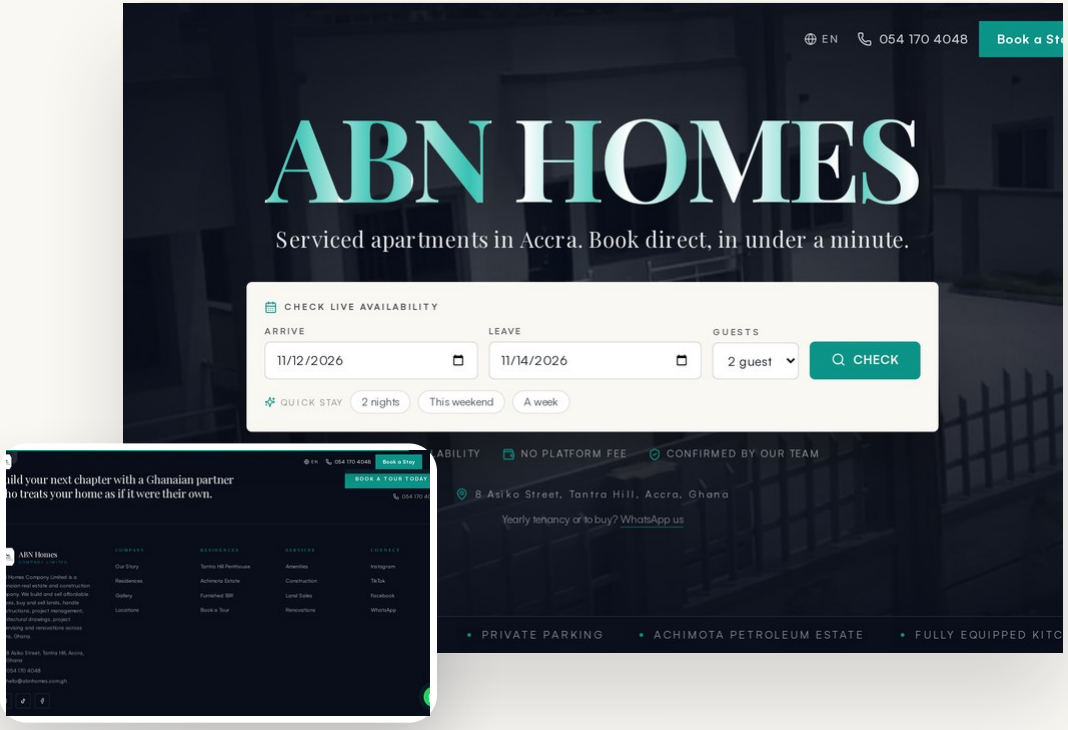
GUEST EXPERIENCE

The brochure is the booking funnel

Every page is server-rendered with per-page metadata, sitemap and robots — the site reads live inventory, so a unit published in the console appears immediately.

Bilingual-ready: English first with an i18n scaffold; amenities, locations, gallery, mortgage calculator and contact channels included.

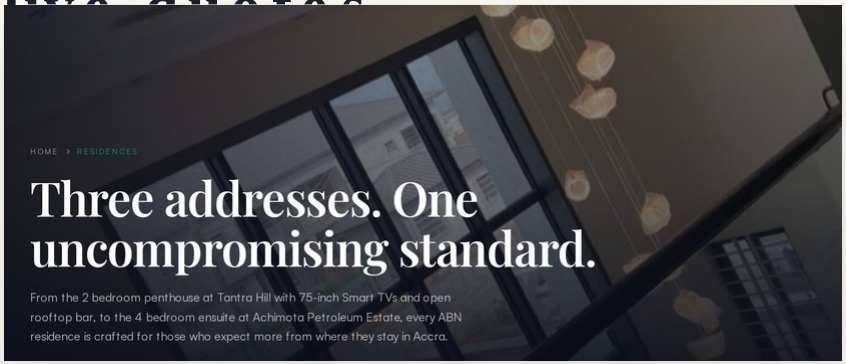
Same design language as this deck: warm cream, deep navy, teal accents.



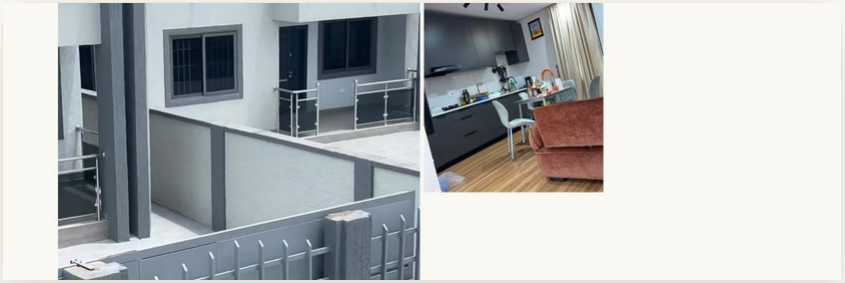
- Live homepage and current-homes grid

GUEST EXPERIENCE

Marketplace: honest listings, live quotes



- Residences grid with filters



- Stay page: gallery, quote, availability, reviews

⇄ **Live pricing** — nightly quotes from active rate plans, never stale

⌕ **Search** — dates, guests, bedrooms, amenities, price

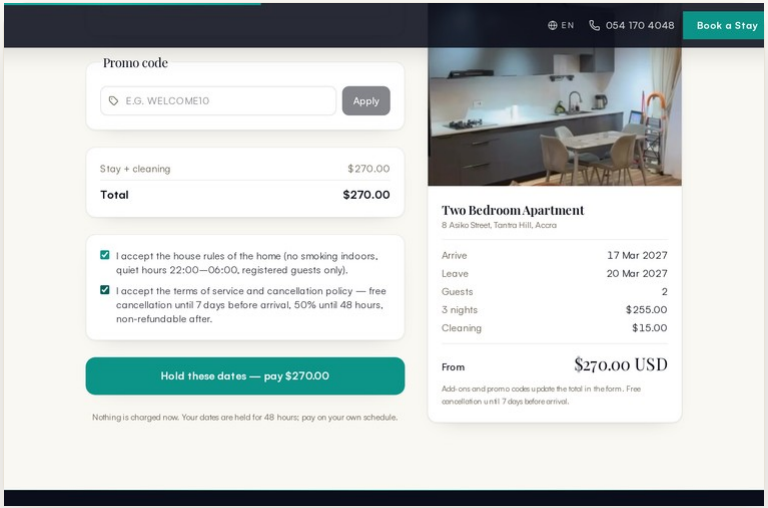
★ **Reviews** — moderated, with per-category sub-scores

♥ **Wishlist** — saved stays for signed-in users

⌘ **Resilient URLs** — slugs backfilled and lookup hardened (fixed 11 Nov)

GUEST EXPERIENCE

Booking: the only public write, the most guarded



- Ninety-second request form with all-inclusive quote

Nothing the browser sends is trusted — price is recomputed, availability re-checked, the unit must be published.

48h hold
dates reserved, booking code issued

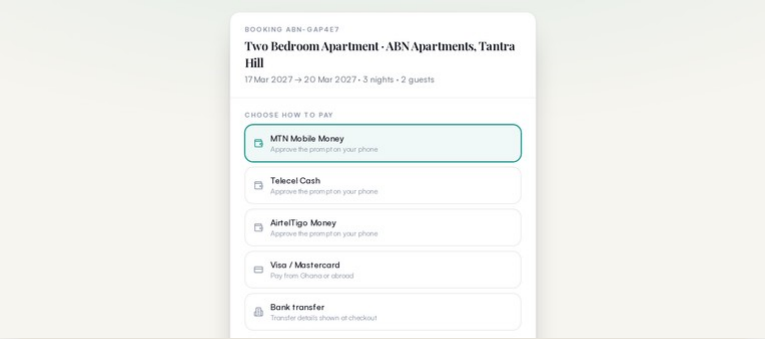
Exclusion constraint
double bookings impossible at the database

In-transaction validation
coupons and add-ons cannot double-apply

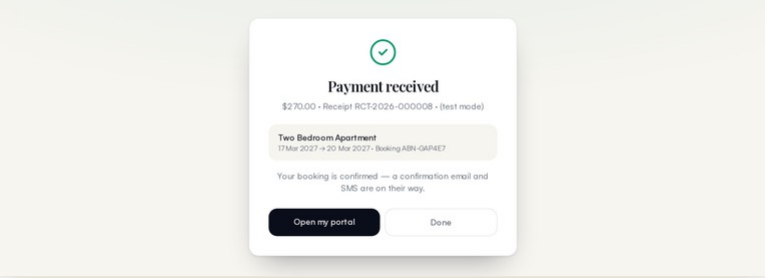
Portal code
guest gets a trip page without an account

GUEST EXPERIENCE

Payments: one code path, two rails



- Checkout: channel, amount, promo code



- Settled: booking auto-confirmed

A gateway abstraction hides Paystack behind an interface. Today the simulation rail runs — same ledger writes, same receipts — so go-live is an env-var change, not a code change.

CHANNELS

MTN MoMo · Telecel Cash · AirtelTigo · Visa / Mastercard · Bank transfer

Webhook signature-verified — unsigned calls get 401

Refunds post reversing entries, payment marked REVERSED

THE WHOLE LOOP

One booking, end to end

1 Discover

marketplace and stay pages with live quotes

2 Hold

48h reservation, booking code, outbox messages queued

3 Pay

simulated MoMo checkout settles into the ledger

4 Confirm

auto-confirmed, receipts + staff alerts fire

5 Stay

trip page with access notes, Wi-Fi, house rules

6 Review

moderated review lands back on the listing

✓ Verified on production 11 Nov: **13/13 assertions passed**, including refund reversal.

CHAPTER 02 / 03

Running the Business

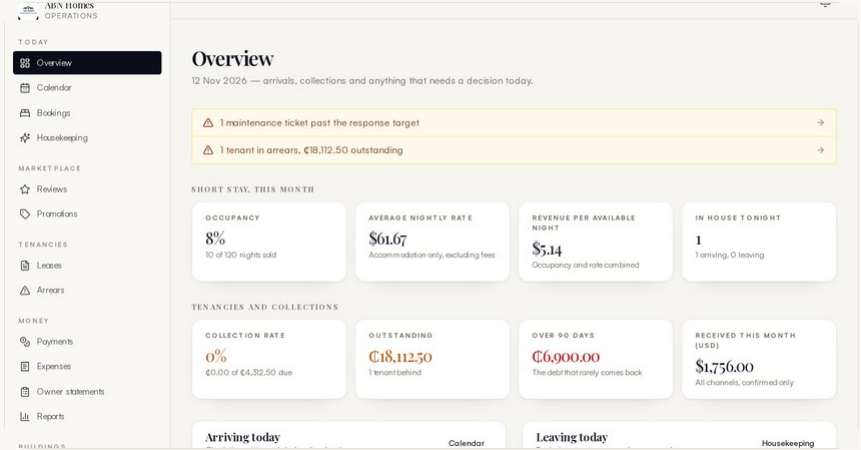
The admin console, the ledger, operations, communications and the portals staff and stakeholders live in.

RUNNING THE BUSINESS

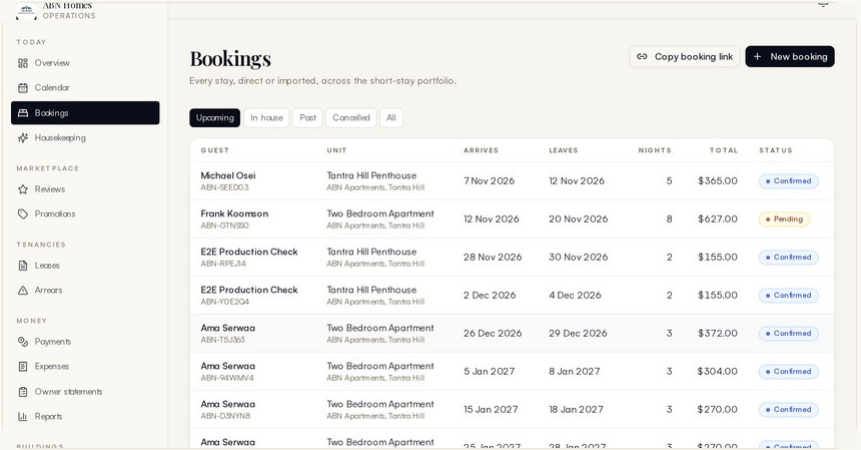
22 pages of operations, one login

Same console, different powers — the role decides the menu.

- 22 pages
- 6 roles from SUPER_ADMIN to VENDOR
- ✓ Every write audited
- Role-gated middleware



Operational snapshot with alerts

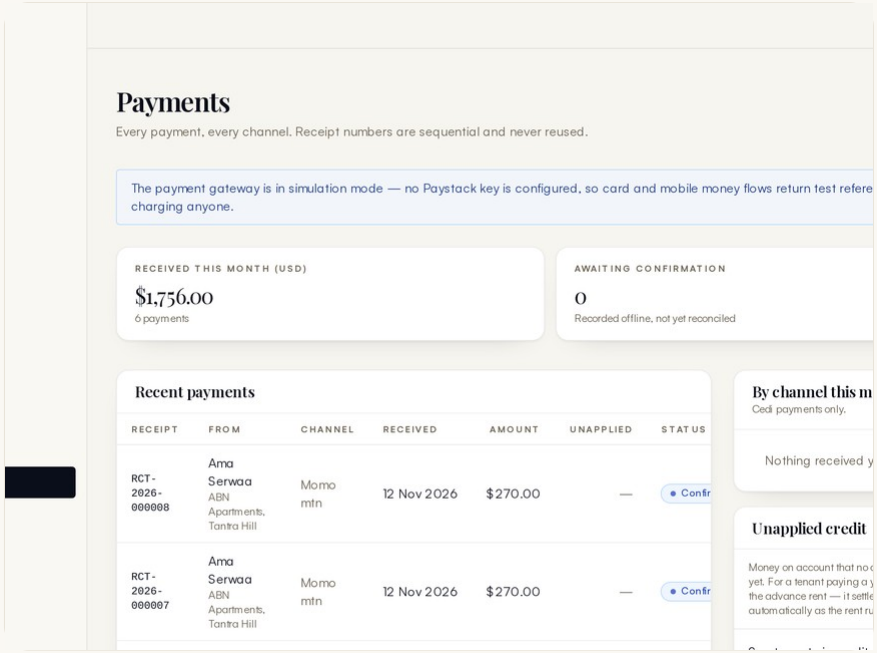


Bookings with balances and status

A housekeeper signs in to housekeeping powers only; finance sees money; SUPER_ADMIN sees all — including user suspension and the audit trail.

RUNNING THE BUSINESS

Every cedi moves through a double-entry ledger



Ledger, payments, expenses and payouts

A payment is a balanced ledger pair allocated across charges — not a status flag. Statements are queries, not reconstructions.

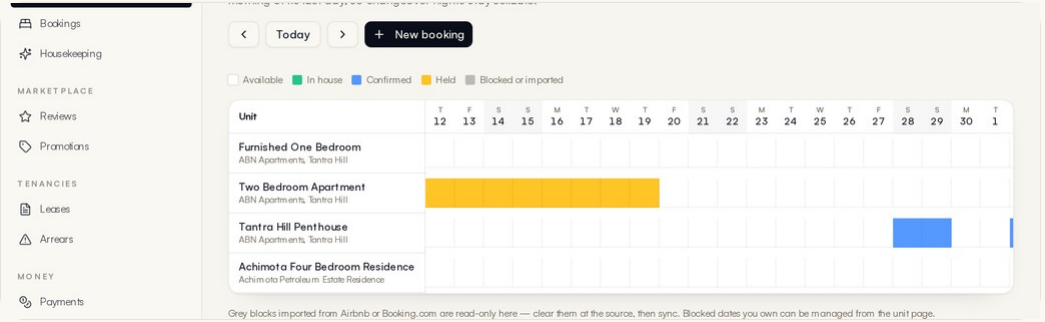
- **Append-only** — ledger edits are blocked by database triggers
- **Reversals** — refunds post reversing entries, never edits
- **Allocations** — payments settle charges oldest-first
- **Statements & payouts** — owner balances always live

RUNNING THE BUSINESS

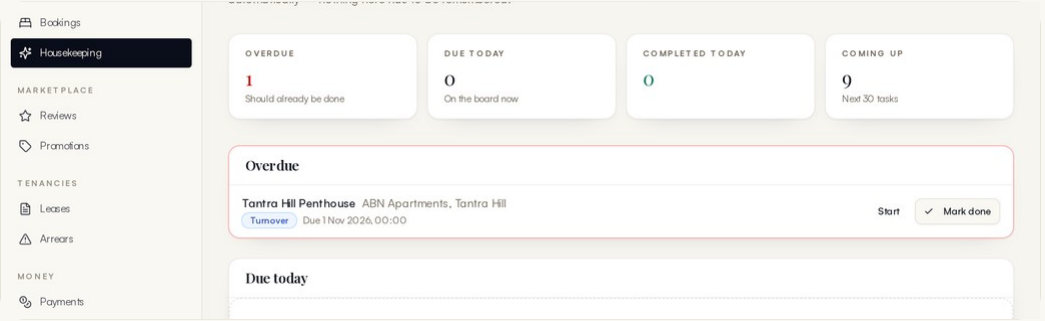
Turnover, maintenance, assets — the daily grind, systematised

From check-out to owner statement, nothing lives in a notebook.

- ✓ Check-out creates the turnover task automatically
- ◔ **Preventive schedules** — generators, AC servicing, pest control
- ⚒ Asset registry ties equipment to units with service history
- $ Costs captured as expenses flow into owner statements



Occupancy calendar



Housekeeping board

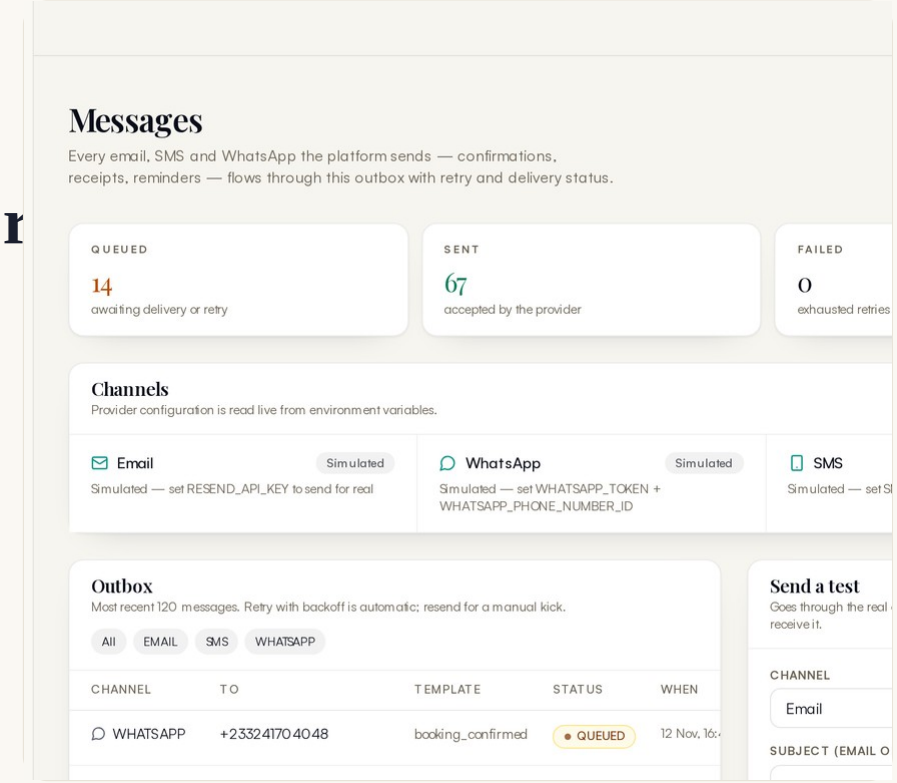
RUNNING THE BUSINESS

An outbox, not a prayer

Every outbound message is queued in the database with channel, status and attempt counter; a cron sweeper drains it. No provider configured yet? Simulation mode records full content and marks sent — the communication design is testable before any sender account exists.

Email · SMS · WhatsApp

Staff bell — in-app notifications with unread counts for booking requests, payments, task assignments, linked to records.

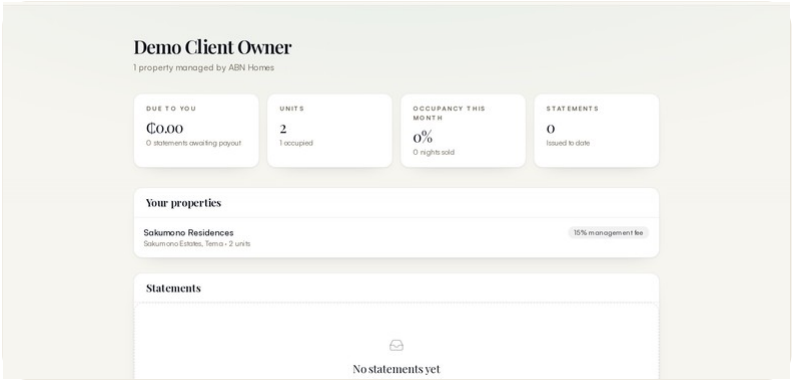


Message log with per-channel status

RUNNING THE BUSINESS

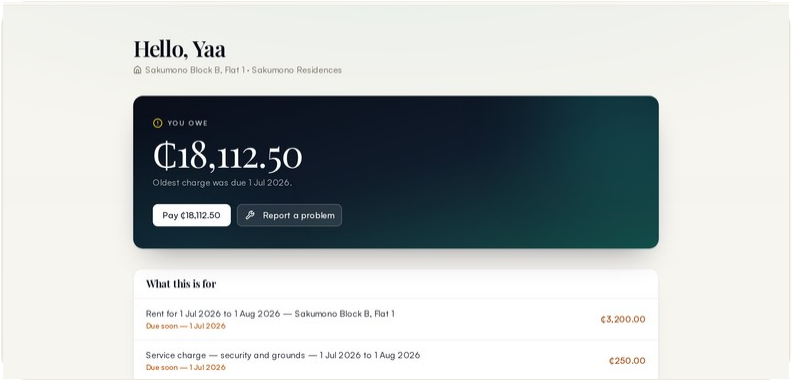
Owners and tenants serve themselves

Two self-service views over the same ledger — statements and requests without a middleman.



Owner portal: performance, statements, payouts

- ✓ Occupancy and revenue per property
- ✓ Expenses charged against their units
- ✓ Payout batches with approval states



Tenant portal: lease, rent account, requests

- ✓ Lease and rent account with arrears position
- ✓ Maintenance request desk with replies
- ✓ Payment history and documents

● Same phone-OTP identity as guests — one system, role-scoped views.

CHAPTER 03 / 03

Platform & Roadmap

What it is built on, what runs unattended, and what stands between the prototype and commercial launch.

0
The Guest Experience

0
Running the Business

03
Platform & Roadmap

PLATFORM

Boring technology, strict guarantees

Next.js 16 + TypeScript

one app for site, console and portals

PostgreSQL 17 on Supabase + Prisma 6

42-table schema, pooled

NextAuth v5

staff password + phone OTP, role matrix

Database guards

exclusion constraint (no double bookings), append-only ledger triggers

Idempotency

double-submitted payments and bookings are harmless

20s transactions

TX_OPTS raised from 5s after a real production timeout (fixed 11 Nov)

PLATFORM

Five jobs run the business overnight

Job	Schedule	What it does
❄ Sweeper	daily + 15-min	expires unpaid holds, drains the outbox
➤ Outbox drain	with sweeper	sends SMS, email, WhatsApp with retries
▤ Rent run	daily 06:00 UTC	invoices every active long-let lease
▦ iCal sync	daily 06:17 UTC	pulls OTA calendars, blocks booked dates
↻ Keep-alive	every 6h	stops Supabase free tier pausing

⚿ Cron routes are bearer-secret guarded — verified 401 on unauthenticated calls.

ROADMAP

Four switches to launch, then growth

⚡ Go-live switches (env vars, no code)

1. **Paystack live keys** — real MoMo and card settlement
2. **SMS sender (Arkesel/Hubtel)** — real OTPs and guest messages
3. **Email sender** — receipts and confirmations
4. **Credential hygiene** — rotate DB password, change seeded staff password, sync CRON_SECRET

↗ Then, in order of value

- → **WhatsApp Business API** for trip updates
- → **Outbound iCal** to Airbnb / Booking.com
- → **Referral rewards activation**
- → **Photo uploads on tasks** ; trend dashboards; image pipeline

Built. Tested. Live.

- ✓ Full platform verification passed 11 Nov 2026 — every page, every flow
- ✓ Production E2E: hold, payment, confirmation, notifications, refund — 13/13
- ✓ Companion deliverables: build documentation (docx), this deck, narrated demo video